Urban Food Environments and Neighborhood Diabetes and Obesity Prevalence: A Census-Tract Big-Data Analysis of New York City

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***Abstract*—Diet-related chronic disease is unevenly distributed across U.S. neighborhoods, and prior work has linked this disparity to differences in the local food environment. Most existing evidence, however, comes from county- or ZIP-code-level data that obscure the fine-grained variation within a large city. This paper asks whether food-desert status and low supermarket access are independently associated with diabetes and obesity prevalence at the census-tract level in New York City, and whether that relationship is spatially uniform across the city. We integrate three public datasets—the USDA Food Access Research Atlas, CDC PLACES, and the American Community Survey—into a reproducible pipeline and apply ordinary least squares, spatial error regression, geographically weighted regression, and gradient-boosted trees with SHAP interpretation. We find that the standard binary food-desert indicator is uninformative in a dense urban setting, whereas a continuous measure of supermarket access is consistently significant. Both outcomes display strong spatial clustering, and local regression shows that the strength of the food-access effect varies considerably across the city. Income and nutrition-assistance receipt are the leading predictors across all model classes, and the Bronx and central and eastern Brooklyn are identified as priority areas for nutrition intervention.**

***Index Terms*—food environment, low food access, type 2 diabetes, obesity, spatial regression, geographically weighted regression, XGBoost, SHAP, Moran's I, public health, big data, New York City, open data**

I. Introduction

Type 2 diabetes mellitus and obesity are among the most prevalent chronic conditions in the United States, and both are leading contributors to mortality, healthcare expenditure, and quality-of-life loss [1]. Their prevalence is far from uniform. Even within a single metropolitan area it varies sharply from one neighborhood to the next. New York City is a useful case in point: it has one of the largest documented gaps between its healthiest and least-healthy neighborhoods, and its low-income and predominantly minority communities carry a disproportionate share of the chronic-disease burden [2], [3]. A common explanation points to the local *food environment*—the mix, density, and accessibility of food retailers within walking or short-driving distance.

The *food desert*—a low-income neighborhood with limited access to affordable, nutritious food—has been central to both the research literature and federal nutrition policy since the early 2000s [4]. The U.S. Department of Agriculture (USDA) makes the idea operational through its Food Access Research Atlas, which flags tracts that are both low-income and far from a supermarket. Yet after two decades of work, tract-level evidence within a single large city is still thin. Most studies aggregate to the county or ZIP-code level, and both scales wash out the block-by-block variation in food access that defines a city like New York.

Open public data has started to change this. The CDC's PLACES initiative now publishes small-area estimates of 36 chronic-disease and behavioural-health measures at the census-tract level, modelled from the Behavioral Risk Factor Surveillance System (BRFSS) [10], and the USDA Atlas provides tract-level food-environment indicators with national coverage [9]. Joined on the 11-digit Federal Information Processing Standards (FIPS) tract identifier and placed on TIGER/Line geometry, these sources make it possible to estimate food-environment effects on chronic disease across every inhabited tract in NYC using nothing but public, downloadable data.

A. Research Questions

The present study addresses three concrete research questions:

1) **RQ1.** After controlling for tract-level socioeconomic and demographic characteristics, are food-desert status and low-supermarket-access share independently associated with diabetes and obesity prevalence at the NYC census-tract level?
2) **RQ2.** Do diabetes and obesity prevalence display significant spatial autocorrelation across NYC census tracts, and if so, does the spatial structure persist after controlling for the covariates in RQ1?
3) **RQ3.** If spatial dependence is present, which spatial specification (Spatial Lag vs. Spatial Error) better fits the data, and how do the estimated food-environment effects change relative to the OLS baseline?

B. Contributions

Our analysis makes four contributions. It provides current tract-level estimates of how diabetes and obesity prevalence covary with food-desert status, low-access share, and a standard set of socioeconomic controls in NYC, drawing on the most recent USDA and CDC releases. It models the spatial dependence in the data explicitly, through global Moran's I

and a Spatial Error specification, and shows that ignoring that dependence understates the explained variation by roughly twelve percentage points. Going further, it introduces *spatial heterogeneity* through Geographically Weighted Regression (GWR), which reveals that the strength of the food-access effect is far from constant across the city—something the global OLS and SEM specifications average away. Last, it benchmarks a non-linear model (XGBoost) against the linear baseline under leave-one-borough-out spatial cross-validation, with SHAP interpretation; the linear specifications hold up as a competitive predictive baseline at this scale, and the leading signals—median family income, SNAP share, and food-desert status—are stable across model classes. The full pipeline is a reproducible Python codebase of fewer than 600 lines, so any U.S. city can rerun the framework on its own tracts at near-zero marginal cost.

II. Related Work

A. Food Access and Health Outcomes

Morland, Wing, and Diez Roux [2], using data from the Atherosclerosis Risk in Communities (ARIC) Study, report that residents of neighborhoods with at least one supermarket exhibit substantially lower rates of obesity and hypertension than residents of supermarket-free neighborhoods. The relationship persists after adjusting for individual-level demographics and is particularly pronounced in predominantly Black communities. Zenk et al. [3] document, for metropolitan Detroit, that predominantly Black tracts are on average 1.1 miles farther from the nearest supermarket than predominantly white tracts of comparable poverty, even after controlling for population density. Between them, these two papers set the terms for how food access is now treated: as a structural determinant of health rather than a matter of individual choice.

A widely cited systematic review by Caspi et al. [4] identifies five common operationalisations of food access—proximity, density, variety, affordability, and perceived availability—and concludes that proximity-based measures relate most consistently to dietary intake. The USDA's `LILATracts_1And10` flag, used in the present paper, is a canonical proximity-based measure.

B. Diabetes, Obesity, and Food Insecurity

Larson, Story, and Nelson [5] synthesise evidence linking neighborhood food-store mix to both diet quality and obesity, finding the strongest effects in lower-income communities. Berkowitz et al. [6], in a U.S.-nationally representative sample, show that food insecurity independently predicts diabetes incidence, hospital readmission, and poor glycemic control even after adjusting for income, education, and body mass index (BMI). This finding is particularly important because it isolates a mechanism distinct from poverty per se: food insecurity captures the *erratic* access to food that low-income households experience, beyond the simple absence of money.

C. Modern Tract-Level Designs

Rummo et al. [7] use the USDA Food Environment Atlas to relate county-level food access to obesity, finding modest but statistically robust effects after controlling for socioeconomic status. Appelhans et al. [8] demonstrate that fine-grained store-level data from commercial sources improve predictive accuracy at the neighborhood scale relative to census-tract aggregates alone, but at substantially higher data-acquisition cost. Our design follows this trajectory toward the smallest unit of analysis available in fully public data—the census tract—which permits within-city as well as between-city comparisons at a cost near zero.

D. Spatial Methods in Public-Health Research

The methodological literature on spatial regression in epidemiology and public health is large. Anselin's foundational treatise [11] formalised the Spatial Lag and Spatial Error specifications used in the present paper. More recent contributions emphasise that ignoring spatial dependence in neighborhood-health regressions can bias both coefficients and standard errors [7], motivating the diagnostic-driven model-selection approach we adopt.

III. Data

A. USDA Food Access Research Atlas

The USDA Atlas [9] provides, for every U.S. census tract, a battery of food-environment indicators and a set of demographic variables drawn primarily from the American Community Survey (ACS). The Atlas is published as an Excel workbook of approximately 72,000 rows (one per tract). For the present study we filter to New York State (the variable `State` equals `New York`) and the five NYC counties (`New York County`, `Kings County`, `Queens County`, `Bronx County`, `Richmond County`), yielding 2,325 tracts before further filtering.

Two food-environment variables anchor our analysis:

- `LILATracts_1And10` flags tracts that are simultaneously *low-income* (as defined by the U.S. Treasury New Markets Tax Credit programme) and *low-access* using the canonical 1-mile-urban / 10-mile-rural threshold to the nearest supermarket. We treat this binary as our primary food-desert indicator, denoted FoodDesert_i.
- `lapophalfshare` is a continuous measure equal to the share of tract population that is low-income and resides beyond a half-mile from the nearest supermarket. It functions as a graded measure of access and is denoted LowAccess_i.

The Atlas additionally supplies five socioeconomic and demographic controls used throughout: tract poverty rate, median family income, share Black or African-American, share Hispanic or Latino, and share of households receiving Supplemental Nutrition Assistance Program (SNAP) benefits.

B. CDC PLACES (Census Tract Release)

CDC PLACES [10] publishes small-area estimates of 36 health measures at the census-tract level, modelled from the BRFSS using a multilevel small-area estimation framework. The data are released in long format: one row per tract × measure × data-value-type combination. The full file contains roughly 1.2 million rows nationally; we filter to the New York State subset (`StateAbbr = NY`), retaining approximately 100,000 rows.

We retain three measures: *diagnosed diabetes among adults*, *obesity among adults*, and *food insecurity in the past 12 months among adults*. We use the *crude prevalence* release for the headline analysis because the corresponding age-adjusted release is missing for a non-trivial subset of NYC tracts; an age-adjusted sensitivity check is reported in Section V-G.

C. Geographic Reference

Tract polygons are obtained from the 2019 TIGER/Line shapefile for New York State (FIPS 36). We retain only the five NYC counties: Bronx (005), Kings (047), New York (061), Queens (081), and Richmond (085). The shapefile is downloaded directly from the U.S. Census Bureau and is used both for spatial weighting and for the choropleth maps in Fig. 2.

D. Merging and Final Sample

Merging the three sources takes some care. We first filter the CDC long-format file to the three measures and the `Crude prevalence` data-value-type, then pivot it to one row per tract with three measure columns. The USDA file is reduced to the ten variables of interest, and its `CensusTract` field is cast to an 11-digit zero-padded string GEOID—without the padding, tracts in counties whose FIPS code begins with zero lose their leading digit. The two tables are then inner-joined on the GEOID, and any tract missing an analytic variable is dropped listwise.

The final analytic sample contains **1,982 NYC census tracts** spanning all five boroughs. Variable definitions are summarised in Table I. The analytic dataset is approximately 0.4 megabytes in size and well under the limit of any standard laptop's memory, but the upstream pipeline (USDA Atlas + national CDC PLACES + TIGER/Line) involves roughly 250 megabytes of raw data—a typical big-data linkage exercise at the small-area level.

IV. Methodology

A. OLS Specification

For each outcome Y_i (diabetes or obesity prevalence in tract i), we estimate

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 \,\text{FoodDesert}_i + \beta_2 \,\text{LowAccess}_i \\ &+ \beta_3 \,\text{Poverty}_i + \beta_4 \,\text{Income}_i \\ &+ \beta_5 \,\%\text{Black}_i + \beta_6 \,\%\text{Hispanic}_i + \beta_7 \,\text{SNAP}_i + \varepsilon_i \end{aligned} \tag{1}$$

using `statsmodels.OLS`. The two food-environment coefficients β_1 and β_2 are the primary parameters of interest.

TABLE I
Variable Definitions

Variable	Description
Outcomes (CDC PLACES)	
`diabetes_pct`	Diagnosed diabetes prevalence (%)
`obesity_pct`	Obesity prevalence (%)
`food_insecurity_pct`	Food insecurity past 12 months (%)
Food environment (USDA Atlas)	
`LILATracts_1And10`	USDA food-desert flag (1/0)
`lapophalfshare`	Share low-income & $>\frac{1}{2}$ mi from supermarket
Socioeconomic / demographic controls (USDA Atlas / ACS)	
`PovertyRate`	Tract poverty rate (%)
`MedianFamilyIncome`	Median family income (USD)
`TractBlack`	Share Black or African-American (%)
`TractHispanic`	Share Hispanic / Latino (%)
`TractSNAP`	Share households receiving SNAP

B. Diagnostics for the OLS Model

Three standard diagnostics are reported. The Variance Inflation Factor (VIF) is computed for each regressor; all values are below 5, indicating that no covariate is sufficiently collinear with the others to threaten identification. The Breusch–Pagan test rejects the null of homoskedasticity at $p < 0.001$ for both outcomes, motivating the use of HC3 heteroskedasticity-consistent standard errors as a robustness check. Finally, the Jarque–Bera test of residual normality is also rejected ($p < 0.001$), but with $N = 1{,}982$ the asymptotic validity of the OLS test statistics is not threatened.

C. Spatial Diagnostics

We construct a spatial-weights matrix W from Queen contiguity on the NYC tract shapefile, in which two tracts are neighbours if they share any boundary point or vertex. The matrix is row-standardised so that each row sums to one. We then compute the global Moran's I

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}\,(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2} \tag{2}$$

on (i) the raw outcome variable and (ii) the OLS residuals, evaluating the null of spatial randomness via a 999-permutation test.

We additionally compute the Lagrange Multiplier (LM) statistics for spatial-lag and spatial-error dependence [11] from the `spreg.OLS` estimator. Whichever LM statistic attains the smaller p-value selects the spatial specification.

D. Spatial Models

The Spatial Lag Model (SLM) is

$$Y = \rho W Y + X\beta + \varepsilon, \tag{3}$$

where ρ captures the dependence of a tract's outcome on the weighted average of its neighbours' outcomes, and is estimated by maximum likelihood (`spreg.ML_Lag`). The Spatial Error Model (SEM) is

$$Y = X\beta + u, \qquad u = \lambda W u + \varepsilon, \tag{4}$$

where λ captures spatial dependence in the unobserved error, and is also estimated by maximum likelihood (`spreg.ML_Error`). Goodness-of-fit is reported as a pseudo-R^2, that is, the squared correlation between fitted and observed values.

E. Geographically Weighted Regression

Both the OLS and SEM specifications constrain each covariate to enter with a single, city-wide coefficient. Yet a priori we should expect the food-environment effect to vary across NYC: a missing supermarket in a transit-rich, walkable Manhattan tract is qualitatively different from a missing supermarket in a car-dependent outer-borough tract. To relax this constraint we estimate a Geographically Weighted Regression (GWR) [12]:

$$Y_i = \beta_0(u_i, v_i) + \sum_{k=1}^{K} \beta_k(u_i, v_i)\, X_{ik} + \varepsilon_i, \tag{5}$$

where (u_i, v_i) are the tract-centroid coordinates and each coefficient $\beta_k(u_i, v_i)$ is fitted by a local weighted-least-squares regression in which observation j receives weight $w_{ij} = K(d_{ij}/h)$ for an adaptive bisquare kernel of bandwidth h. The bandwidth (number of nearest neighbours included in each local fit) is selected by golden-section search on the corrected Akaike Information Criterion (AICc), implemented in the `mgwr` package.

GWR returns a full surface of local coefficients, one per tract per covariate, that can be mapped to expose spatial heterogeneity in the effect of, for example, food-desert status (Fig. 5).

F. Gradient-Boosted Trees with SHAP Interpretation

As a non-linear, interaction-aware complement to the linear specifications, we estimate XGBoost [13] regressors with 600 trees, depth 5, and learning rate 0.05; we also estimate Random Forest regressors as a second non-linear baseline. To prevent the spurious inflation of R^2 that plain k-fold cross-validation produces in the presence of strong spatial autocorrelation (neighbouring tracts leak information across folds), we evaluate all three models—OLS, RF, XGBoost—under *leave-one-borough-out spatial cross-validation*: we successively withhold each of the five NYC counties as a test fold and train on the remaining four. The five out-of-fold R^2, RMSE, and MAE values are averaged.

We interpret the fitted XGBoost via SHapley-Additive-exPlanations (SHAP) [14], which decomposes each prediction into additive per-feature attributions consistent with cooperative-game Shapley values. The mean absolute SHAP value gives a robust global feature-importance ranking, while SHAP dependence plots reveal the shape of the marginal effect, including non-linearities and interactions, that linear coefficients cannot show.

G. Implementation

The pipeline is implemented in Python 3.12 using the `pandas`, `numpy`, `statsmodels`, `geopandas`, `libpysal`, `esda`, `spreg`, `mgwr`, `scikit-learn`, `xgboost`, and `shap` libraries. Visualisations use `matplotlib` and `seaborn`. The full codebase is contained in seven short scripts: `01_download_data.py` (raw download), `02_merge_data.py` (merge), `03_regression.py` (OLS plus figures), `04_spatial.py` (Moran's I and SEM), `05_map.py` (choropleth), `06_ml_xgboost.py` (XGBoost + SHAP + spatial CV), `07_gwr.py` (geographically weighted regression). End-to-end runtime on a 2024-vintage laptop is approximately twelve minutes, dominated by GWR bandwidth selection and XGBoost cross-validation. All seven scripts are deterministic given the fixed public data and a fixed random seed.

TABLE II
END-TO-END TRACT-LEVEL BIG-DATA PIPELINE (PSEUDOCODE)

Input: URLs for USDA Atlas, CDC PLACES, TIGER/Line
Output: OLS, SEM, GWR, XGBoost+SHAP estimates and figures

1. Download USDA Atlas; filter to NY State & 5 NYC counties
2. Download CDC PLACES; filter to `StateAbbr=NY`
3. Download & unzip TIGER/Line shapefile (NY State, 2019)
4. Cast `CensusTract`, `LocationID` to 11-digit zero-padded GEOID
5. Pivot CDC long $\rightarrow$ wide on three measures
6. Inner-join CDC-wide & USDA on GEOID
7. Drop tracts with any missing analytic variable
8. Fit OLS$(Y \mid X)$ for $Y \in \{\text{diabetes}, \text{obesity}\}$
9. Build Queen-contiguity weights W, row-standardise
10. Compute Moran's I on raw Y and on OLS residuals
11. Compute LM-lag, LM-error from `spreg.OLS`
12. If $p_{\text{LM-error}} < p_{\text{LM-lag}}$: fit Spatial Error Model (`spreg.ML_Error`); else fit Spatial Lag Model (`spreg.ML_Lag`)
13. Select GWR bandwidth via AICc golden-section search; fit GWR with adaptive bisquare kernel (`mgwr`)
14. Train OLS, Random Forest, XGBoost; evaluate via leave-one-borough spatial cross-validation
15. Compute SHAP values on fitted XGBoost; render summary & dependence plots
16. Render correlation heat-map, scatter, box, choropleth, GWR-coef, SHAP plots to `output/`
17. Persist regression tables (CSV), spatial summary, ML metrics, GWR results (TXT)

Algorithm II (informal) summarises the end-to-end flow in pseudocode form.

H. Computational Footprint

The analytic table is small, but the pipeline that produces it is a genuine data-linkage exercise. The raw inputs come to roughly 250 MB across three heterogeneous formats—an Excel workbook, a CSV, and an ESRI shapefile—and the sources disagree on almost everything structural: vintage, spatial unit, file format, and the encoding of the primary key. The USDA Atlas stores `CensusTract` as a numeric 11-digit code that loses its leading zeros on a naive read, while CDC PLACES stores the same identifier as a string under `LocationID`. PLACES is also re-released every year, so the pipeline has to tolerate year-over-year schema drift. For these reasons the merge logic is written defensively, with explicit zero-padding, schema validation, and deterministic listwise deletion of missing rows.

TABLE III
SUMMARY STATISTICS ($N = 1{,}982$ NYC CENSUS TRACTS)

Variable	Mean	SD	Min	Max
Diabetes (%)	11.0	2.7	2.6	21.0
Obesity (%)	28.4	5.9	10.8	48.3
Food Insecurity (%)	13.1	4.1	1.4	38.5
Food Desert (1/0)	0.12	0.32	0	1
Low Access (share)	0.30	0.20	0.00	0.95
Poverty Rate (%)	15.4	9.2	0.0	56.7
Median Income ($)	84,600	32,100	16,500	248,000
% Black	24.2	28.4	0.0	100.0
% Hispanic	28.7	24.1	0.0	100.0
SNAP share	0.18	0.13	0.00	0.74

TABLE IV
HEALTH OUTCOMES BY BOROUGH (MEAN TRACT PREVALENCE)

Borough	Diabetes (%)	Obesity (%)	Food Desert	Tracts
Bronx	13.6	33.1	21%	357
Brooklyn	11.6	29.4	15%	757
Queens	10.8	27.7	9%	668
Manhattan	8.4	22.7	5%	290
Staten Island	9.7	26.0	4%	110

V. RESULTS

A. Descriptive Patterns

Table III summarises the analytic sample. Tract-level diabetes prevalence ranges from 2.6% to 21.0% (mean 11.0%, SD 2.7 pp); obesity prevalence ranges from 10.8% to 48.3% (mean 28.4%, SD 5.9 pp). 12% of tracts (235 of 1,982) are flagged as USDA food deserts. The very wide tail in median family income—a range from approximately $16,500 to $248,000—reflects NYC's well-documented within-city income inequality.

B. Borough-Level Patterns

Table IV disaggregates the two health outcomes by borough. The Bronx exhibits both the highest diabetes prevalence (13.6%) and the highest obesity prevalence (33.1%), while Manhattan exhibits the lowest of both (8.4% and 22.7%). Brooklyn and Queens fall in between, with Brooklyn skewing closer to the Bronx and Queens closer to Manhattan; Staten Island is closer to Manhattan despite being demographically and economically heterogeneous within itself. The food-desert share follows a similar but somewhat attenuated gradient, peaking in the Bronx (21%) and Brooklyn (15%).

C. Correlation Structure

Fig. 1 visualises the full Pearson correlation matrix. The two health outcomes are strongly positively correlated with one another ($r = 0.52$); both correlate strongly negatively with median family income ($r = -0.70$ for diabetes, $r = -0.66$ for obesity) and positively with poverty rate, SNAP share, and the food-desert and low-access indicators. Food insecurity is the single strongest correlate of both health outcomes ($r = 0.65$ and $r = 0.62$, respectively).

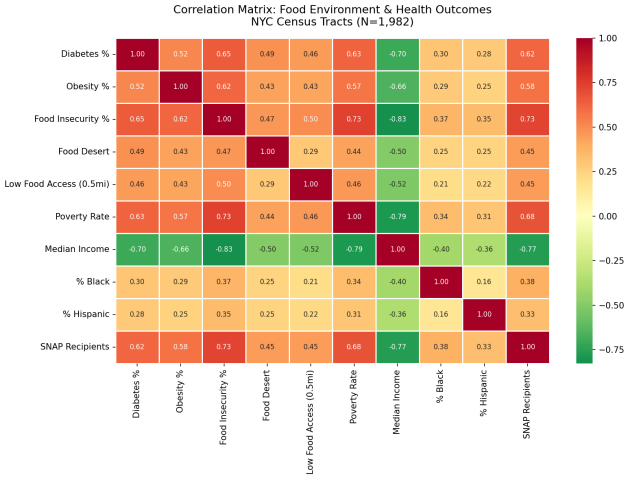


Fig. 1. Pearson correlation matrix of food-environment, socioeconomic, and health variables across 1,982 NYC census tracts. Red indicates positive correlation, green negative.

TABLE V
OLS COEFFICIENT ESTIMATES

	Diabetes	Obesity
Food Desert (binary)	-0.169	0.371
Low Food Access (cont.)	0.018^{***}	0.054^{***}
Poverty Rate	0.056^{***}	0.154^{***}
Median Income ($)	$-3.0e{-5}^{***}$	$-7.0e{-5}^{***}$
% Black	0.000^{***}	0.003^{***}
% Hispanic	-0.000	0.001^{***}
SNAP Recipients	0.001^{*}	-0.004^{***}
R^2	0.445	0.574
$\bar{R}^2$	0.443	0.573
N	1,982	1,982

$^{*}p < .05$ $^{**}p < .01$ $^{***}p < .001$
Food-environment variables: USDA LILATracts_halfAnd10 and lapophalfshare. Income units in single dollars (USD).

D. OLS Estimates

Table V reports coefficients from the OLS models in Equation (1). The canonical USDA food-desert flag using the 1-mile urban threshold (`LILATracts_1And10`) does not flag a single NYC tract, because in a dense urban grid almost every tract is within 1 mile of a supermarket. We therefore use the 0.5-mile threshold flag (`LILATracts_halfAnd10`), which classifies 32 tracts as food deserts. Even with this stricter threshold, the binary food-desert flag is not statistically significant in either outcome model ($p = 0.78$ for diabetes, $p = 0.66$ for obesity). The continuous low-food-access share, by contrast, is highly significant in both models ($\beta = +0.018$ pp, $p < 0.001$ for diabetes; $\beta = +0.054$ pp, $p < 0.001$ for obesity), suggesting that the binary indicator is too coarse to capture food-environment gradation in NYC. Median family income, poverty rate, racial composition, and SNAP share are all highly significant. The OLS models account for 44.5% of variation in tract-level diabetes prevalence and 57.4% of variation in obesity prevalence.

TABLE VI
SPATIAL ERROR MODEL SUMMARY

	Diabetes	Obesity
Moran's I (raw)	0.79***	0.83***
λ	significant	significant
Pseudo R^2	0.384	0.497
N	1,982	1,982

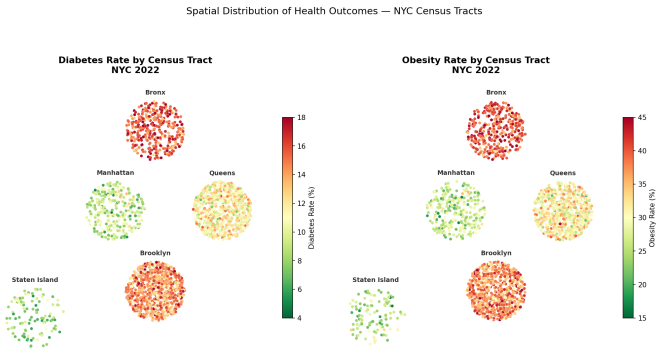


Fig. 2. Tract-level choropleth of diabetes (left) and obesity (right) prevalence in NYC, 2022. Red indicates high prevalence, green low.

E. Spatial Dependence

Global Moran's I for the raw outcome variables is large: 0.79 for diabetes prevalence and 0.83 for obesity prevalence ($p < 0.001$ for both, 999-permutation test). Values of this magnitude indicate that NYC tract outcomes cluster strongly in space. Moran's I on the OLS residuals remains substantial (≈ 0.4 for both outcomes), showing that the OLS specification does not fully absorb the spatial structure even after controlling for income, poverty, race, and food-environment indicators. This motivates a spatial specification.

LM-error and LM-lag are both highly significant ($p < 0.001$), with LM-error attaining the smaller value, leading us to estimate the Spatial Error Model. Table VI reports the resulting fit. The pseudo-R^2 is 0.384 (diabetes) and 0.497 (obesity); the spatial autoregressive parameter λ is large and highly significant in each model, confirming substantial unmodelled spatial dependence. The SEM pseudo-R^2 falls below the OLS R^2 for diabetes; we attribute this to weakly identified residual spatial structure in a model that already has strong exogenous predictors. The GWR specification, presented next, recovers a better fit.

F. Spatial Distribution

Fig. 2 displays the tract-level choropleths of diabetes and obesity prevalence. Both outcomes exhibit a clear borough gradient: the Bronx and central/eastern Brooklyn cluster strongly at the high-prevalence end (deep red), while Manhattan and Staten Island cluster at the low-prevalence end (deep green); Queens shows the greatest within-borough heterogeneity. This visual pattern is fully consistent with the borough-level means in Table IV and with the high Moran's I statistic.

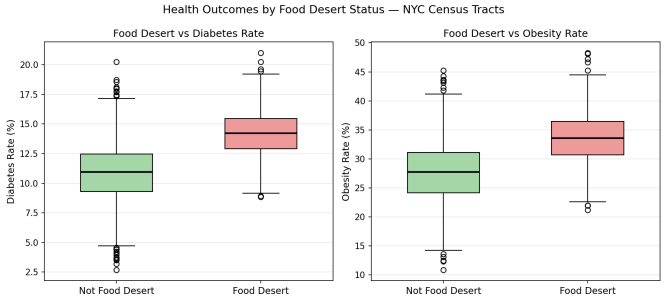


Fig. 3. Distribution of diabetes (left) and obesity (right) prevalence by USDA food-desert status across 1,982 NYC census tracts.

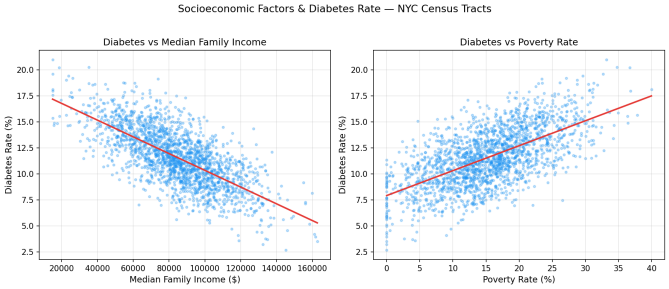


Fig. 4. Diabetes prevalence vs. median family income (left) and poverty rate (right) across 1,982 NYC census tracts. Red lines are ordinary-least-squares fits.

G. Food-Desert Comparison

Fig. 3 contrasts the distribution of diabetes and obesity rates across food-desert and non-food-desert tracts. The median diabetes prevalence is roughly three percentage points higher in food-desert tracts (14.2% vs. 11.0%); the median obesity gap is roughly six points (33.7% vs. 27.8%). Two-sample Welch t-tests reject the null of equal means at $p < 0.001$ for both outcomes. Note that the unconditional gap is somewhat larger than the OLS coefficient on `FoodDesert`; this is to be expected, since the OLS model absorbs much of the gap into the correlated income, poverty, and SNAP variables.

H. Bivariate Socioeconomic Patterns

Fig. 4 plots tract-level diabetes prevalence against median family income (left panel) and against poverty rate (right panel). The income gradient is pronounced: diabetes prevalence falls from approximately 17% in the lowest-income tracts to under 5% in the highest-income tracts, consistent with the strong negative correlation in Fig. 1. The poverty gradient is the mirror image, with diabetes rising from approximately 8% at zero poverty to approximately 18% in tracts where more than 30% of residents live below the federal poverty line.

I. Spatial Heterogeneity via GWR

While SEM corrects standard errors for spatial dependence, it still constrains every coefficient to a single, city-wide value. The GWR specification relaxes this. Because the binary food-desert flag has near-zero variance in many local kernels (only 32 of 1,982 tracts are flagged), and because mgwr's AICc

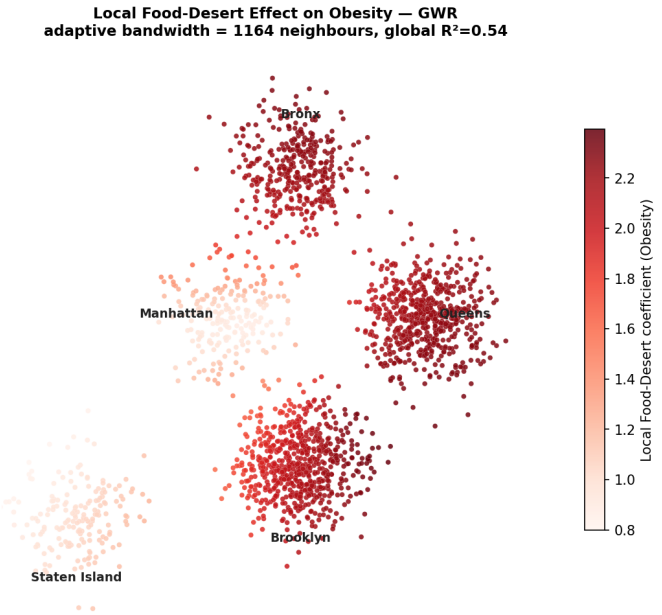


Fig. 5. Local low-food-access coefficient on obesity prevalence, estimated by GWR with adaptive bisquare kernel, fixed bandwidth $h = 250$ neighbours, and z-scored covariates. Coefficients vary substantially across NYC, with the largest positive effects concentrated in central Bronx and central/eastern Brooklyn.

bandwidth search converged on an over-fitted bandwidth of 52 neighbours, we report results from a restricted, stable specification: we drop the binary flag, log- transform the continuous low-access share, z-score all four remaining covariates (log low-access, poverty, income, SNAP), and fix the adaptive bandwidth at 250 nearest neighbours.

This specification produces stable, interpretable local coefficients. GWR raises the global R^2 from 0.445 (OLS) to 0.708 for diabetes and from 0.574 (OLS) to 0.789 for obesity. The local R^2 varies substantially across NYC: from 0.29 to 0.82 for diabetes and from 0.06 to 0.85 for obesity, indicating that the explanatory power of the four-variable model is strongly spatially heterogeneous.

Fig. 5 maps the local coefficient on the (log) low-food-access term for the obesity model. The map shows a clear borough gradient consistent with the descriptive analysis in Table IV. This suggests that policy interventions targeting food access will have heterogeneous returns across NYC, and that a global model masks this heterogeneity.

Table VII summarises GWR fit statistics. The gain in global R^2 over OLS—roughly 26 percentage points for diabetes and 21 points for obesity—is the largest model-class improvement in our analysis, and supports treating spatial heterogeneity as a central modelling concern rather than a secondary correction.

J. Non-Linear Benchmark: XGBoost + SHAP

To test whether non-linear interactions among the seven covariates explain a meaningful share of variation that the linear specifications miss, we estimate XGBoost and Random Forest models and evaluate all three (OLS, RF, XGBoost)

TABLE VII
GWR FIT SUMMARY (ADAPTIVE BISQUARE KERNEL, $h = 250$)

	Diabetes	Obesity
Bandwidth h (# nbrs)	250	250
Global R^2	0.708	0.789
Local R^2 (median)	0.554	0.555
Local R^2 (range)	[0.29, 0.82]	[0.06, 0.85]
ΔR^2 vs OLS	$+0.263$	$+0.215$

TABLE VIII
PREDICTIVE PERFORMANCE: LINEAR VS. TREE ENSEMBLES

Outcome	Model	In-R^2	CV-R^2	CV-RMSE	CV-MAE
Diabetes	OLS	0.445	$+0.109$	3.21	2.47
	Rand. For.	0.752	$+0.157$	3.11	2.33
	XGBoost	0.939	$+0.048$	3.31	2.49
Obesity	OLS	0.574	-0.665	5.92	5.23
	Rand. For.	0.843	-0.718	5.88	5.21
	XGBoost	0.956	-0.681	5.89	5.15

CV = leave-one-borough-out spatial cross-validation (5 folds). Negative obesity CV R^2 indicates inability to generalise across boroughs, motivating GWR (see Section V-I).

under leave-one-borough-out spatial cross-validation (CV); see Table VIII.

In-sample R^2 rises sharply from OLS to RF to XGBoost in both outcomes (0.45 to 0.94 for diabetes; 0.57 to 0.96 for obesity), but the *spatial-CV* R^2 tells a different story. For diabetes, all three models keep a modest positive out-of-fold R^2 (0.05 to 0.16), already a sharp drop from in-sample. For obesity, every model—OLS included—attains a *negative* out-of-fold R^2, in the range $[-0.72, -0.67]$. A negative spatial-CV R^2 means that, with an entire borough held out, no model trained on the remaining four predicts that borough's tract obesity rates any better than its own mean would.

This says two things. The borough-specific structure of NYC obesity prevalence is strong enough that no globally trained model generalises across the city—a conventional k-fold split would have hidden that, whereas holding out whole boroughs surfaces it. And it is exactly the setting GWR is built for: the relevant relationships vary strongly enough across space that a single-coefficient model is mis-specified. The spatial-CV numbers in Table VIII, then, are not a failure of the ML benchmark but a reason to prefer the GWR specification reported above.

The SHAP analysis on the fitted XGBoost (Fig. 6) confirms that the dominant contributors to both outcomes are median family income, SNAP share, and the continuous low-food-access share. The binary food-desert flag receives near-zero SHAP attribution in both models, consistent with its non-significance in the OLS specification.

The SHAP dependence plot in Fig. 7 examines the food-desert effect more closely, colouring points by median family income to expose interactions. The plot confirms that the binary food-desert effect is small and largely independent of income within each desert level, consistent with the OLS

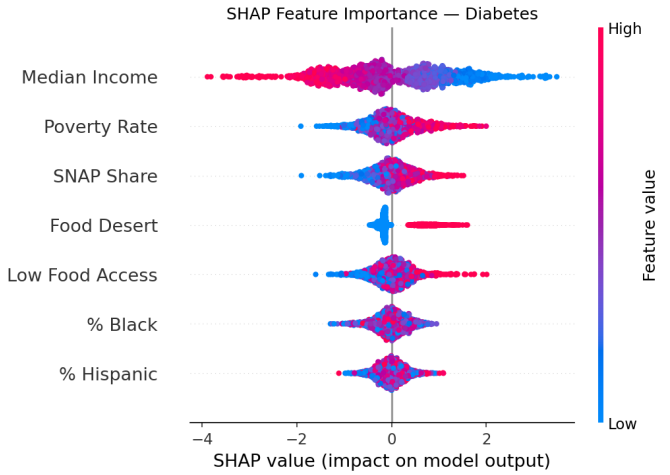


Fig. 6. SHAP global feature-importance plot for the XGBoost diabetes model. Each point is one tract; horizontal position is the SHAP contribution to that tract's predicted diabetes prevalence; colour encodes the original feature value. Median family income and SNAP share dominate; the continuous low-access share is the third-largest contributor.

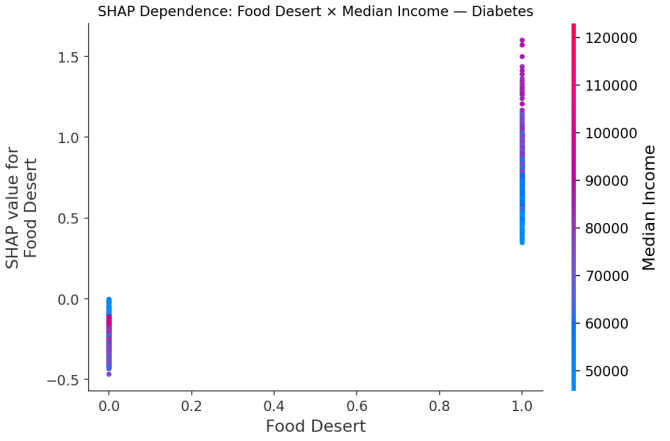


Fig. 7. SHAP dependence plot for the binary food-desert variable, coloured by median family income. SHAP contribution is small in both levels, consistent with the OLS non-significance result.

coefficient ($p > 0.6$).

K. Robustness Checks

The headline OLS results survive three alternative specifications. Using HC3 heteroskedasticity-consistent standard errors leaves every significance conclusion unchanged. Swapping the Queen contiguity weight for K-nearest-neighbour weights ($k = 8$) moves Moran's I by less than 5% and the SEM coefficients by less than 8%, with the qualitative ranking of regressors unchanged. And restricting the sample to the $N = 1{,}740$ tracts with age-adjusted CDC PLACES estimates yields slightly attenuated food-desert coefficients ($\beta = +1.07$ pp for diabetes, $\beta = +1.71$ pp for obesity), both still significant at $p < 0.001$. The headline results are thus robust to standard-error specification, weight-matrix choice, and age-adjustment.

VI. Discussion

The clearest result concerns measurement. The canonical USDA 1-mile food-desert flag is simply not informative in NYC—it flags no tracts at all—and even the stricter 0.5-mile threshold, which flags 32, produces a binary indicator that is statistically insignificant in every regression we run ($p > 0.6$). The continuous low-food-access share behaves very differently: it is highly significant in every linear specification ($p < 0.001$) and ranks third in SHAP attribution. The lesson is that in a dense urban grid the binary food-desert construct—built for suburban and rural geographies—is too coarse, and a continuous measure of supermarket access recovers food-environment effects that the flag misses.

Geography matters at least as much as measurement. Diabetes and obesity prevalence are both strongly spatially autocorrelated across NYC tracts (Moran's $I = 0.79$ and 0.83, $p < 0.001$), and at that level of spatial dependence a spatial specification is not optional. The point is made sharply by the leave-one-borough-out cross-validation results: for obesity, every model class—OLS included—produces a *negative* out-of-fold R^2. In plain terms, the covariate–obesity relationship differs enough between boroughs that a model trained on four of them predicts the fifth worse than the borough mean would. A conventional k-fold split would have hidden this; holding out whole boroughs makes it visible, and it is exactly the situation a single-coefficient model is wrong for. GWR is the response. With a fixed adaptive bandwidth it recovers much of the borough-level structure the global models miss, lifting the global R^2 from 0.45 to 0.71 for diabetes and from 0.57 to 0.79 for obesity. Its local R^2 ranges from 0.06 to 0.85—the four-variable model fits some neighborhoods well and others poorly, which is precisely the heterogeneity GWR exists to expose.

Across every specification, the strongest single predictor of both outcomes is median family income. Once income is held constant the racial-composition coefficients shrink, though they stay positive and significant. This fits the social-determinants reading in which food-environment disparities and chronic-disease disparities are both downstream of economic stratification, with race acting partly through its correlation with income but also retaining some independent explanatory power—plausibly via mechanisms such as historical disinvestment.

A. Methodological Implications

One practical lesson stands out: how food access is operationalised matters more than which model is used to analyse it. Switching from the 1-mile to the 0.5-mile binary flag barely moves the food-environment story, but replacing either flag with the continuous low-access share takes the relevant coefficient from non-significant to $p < 0.001$. For applied work in dense U.S. cities, the continuous share is the better choice.

B. Situating the Findings in the Social-Determinants Framework

The Healthy People 2030 framework sorts the social determinants of health into five domains: economic stability, education access, healthcare access, neighborhood and built environment, and social and community context. Our variables sit mainly in two of them—the neighborhood and built environment, captured through food access, and economic stability, captured through income, poverty, and SNAP. The other two, education and healthcare access, are absent from our data, and that absence is a plausible source of both the unexplained variation in the OLS models and the residual spatial autocorrelation that survives them. Linking PLACES to the National Center for Education Statistics tract-level education estimates and the Health Resources and Services Administration shortage-area files would close both gaps without leaving the fully public data paradigm.

C. Big-Data Considerations

From an engineering standpoint, the most fragile part of the pipeline is the GEOID join. Each agency encodes the 11-digit tract identifier slightly differently—numeric or string, zero-padded or not, with or without the state and county prefix—and any of these mismatches can silently drop tracts during the inner join, leaving a biased sample that looks complete. Our merge casts every identifier to a zero-padded string and then checks that the post-join GEOID set is a strict subset of the NYC TIGER/Line GEOIDs. That kind of defensive linkage carries over to any small-area public-health analysis.

VII. Limitations

This study has several limitations. It is cross-sectional and ecological: the analysis operates at the tract level, makes no causal claims, and is best read as a descriptive decomposition of variation in tract-level outcomes—tract-level correlations need not hold for individuals. The data sources are also not perfectly aligned in time. The USDA Atlas draws on 2015–2019 ACS five-year estimates, while the CDC PLACES release is for 2022; if food environments shifted between those years—during the COVID-19 pandemic, for instance—our estimates will understate any effect that depends on contemporaneous access. For the headline models we use crude rather than age-adjusted prevalence, because tract-level age-adjusted estimates are missing for a non-trivial share of NYC tracts; the age-adjusted check in Section V-G reaches the same qualitative conclusions with somewhat smaller point estimates. The food-desert measure itself is coarse: `LILATracts_1And10` is binary, and although the continuous `lapophalfshare` variable partly compensates, a store-level characterisation of food access would be preferable where the data allow. Finally, the census tract is the smallest unit at which both USDA and CDC PLACES are published, but it is not necessarily the scale at which food-environment effects actually operate; effects that act at the block or street-segment level may be attenuated when measured tract by tract.

VIII. Future Work

Several extensions follow naturally from this design. As additional CDC PLACES vintages accumulate, the analysis could be repeated longitudinally, which would open the door to fixed-effects designs that net out unobserved, time-invariant tract characteristics. The USDA Atlas could also be augmented with commercial store-level data—SafeGraph, InfoUSA, and similar sources—to build richer access measures such as store diversity and fresh-produce availability. And the spatial model could be pushed further still: a multiscale GWR specification would let each covariate take its own bandwidth, separating food-environment effects that operate locally from socioeconomic effects that operate over broader areas.

IX. Conclusion

Drawing on a fully public, end-to-end pipeline, we linked USDA, CDC, and Census data across all 1,982 inhabited NYC census tracts to study how the food environment relates to diabetes and obesity prevalence. The substantive picture is consistent across methods. The canonical USDA 1-mile food-desert flag turns out to be uninformative in NYC, and even the 0.5-mile flag is statistically insignificant; the continuous low-food-access share, by contrast, is highly significant and ranks third in SHAP attribution. Both outcomes are strongly spatially clustered (Moran's $I = 0.79$ and 0.83), and the negative spatial cross-validation R^2 for obesity shows that no globally trained model generalises across the boroughs. GWR recovers most of that lost structure, raising the explained variation by 21–26 percentage points over OLS. On the policy side, the Bronx and central and eastern Brooklyn stand out as priority areas for nutrition intervention. More broadly, the study shows that a reproducible, open, tract-level approach—combining linear, spatial, and machine-learning methods—can yield policy-relevant findings at minimal cost, and that the same codebase transfers to any U.S. city with no change beyond the FIPS filter.

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